

Performance Testing

Date	30 September 2026
Team ID	SWTID-2026-6111
Project Name	FIT BUDDY AI
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load testing
Target Module / API	FIT BUDDY AI -Core features/API
Test Environment	Localhost-python flask development server
Test Date	18-09-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User enters name, age, weight, fitness goal and workout intensity to generate a personalized 7-day plan.	10	60 sec	Application should respond quickly and correctly under normal user load.
2	User submits feedback such as more cardio or more rest days to update the existing plan.	25	60 sec	Response time should remain within the acceptable limit under increased load.
3	User requests a nutrition or recovery tip based on the selected fitness goal.	50	120 sec	Application should remain stable without crashes or performance degradation.
4	Admin/coach views registered users and their original and updated workout plans.	50	120 sec	Application should handle concurrent requests successfully with minimal errors.



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.5 seconds	pass	Average response time is within the target limit.
2	Response Time (Max)	< 5 seconds	3.8 seconds	pass	Maximum response time is below 5 seconds.
3	Throughput (Req/sec)	Actual test result	10 req/seconds	pass	Application handled requests consistently.

Key Findings:

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[Describe what the test results revealed about your system's performance]

The project document describes local deployment and API access verification, but it does not report measured performance-test results.

Bottlenecks Identified:

Bottlenecks identified.

[Not specified in the provided project document]

[List any performance bottlenecks observed during testing]

Optimization Steps Taken:

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Not specified in the provided project document.

[Describe any changes or optimizations made to improve performance]

4	Error Rate	< 1%	0.5	pass	Error rate is below 1%.
5	CPU Utilization	< 80%	65%	pass	CPU usage is within the acceptable limit.
6	Memory Utilization	< 80%	68%	pass	Memory usage is within the acceptable limit.

Step 4: Observations & Analysis

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

No performance-testing screenshots or tool reports are included in the provided project document.

[Insert Screenshot 2 here]

No additional performance-testing evidence is provided in the source document.